



Negation Origin Tracing

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The Problem “Not” is Hard

Input Sentence	Logic/Ground Truth	Model Prediction	Outcome
"The movie was good"	Positive	Positive	✓ Success
"The movie was not good"	Negative	Positive	✗ Failure
"The capital of Cuba is"	Havana	Havana	✓ Success
"The capital of Cuba is not"	Not Havana	Havana	✗ Failure

- ❑ LMs are good at association but struggle with logical negation
- ❑ Models often ignore the negation and relies on the strong association between subject and object
- ❑ LLMs mishandle negation (“not good” ≠ “good”)
- ❑ Unclear where this failure occurs
- ❑ Impacts core tasks like sentiment and reasoning



Overview

01

Locating Negation:

- Identify which layers and neurons encode it

02

Validate Casualty:

- Confirm if those features drive model predictions

03

Trace the Negation

- When does the negation become meaningful?

Methodology

Models & Input

- ❑ **Models:** DistilBERT pre-finetuned on SST-2 (91% accuracy)
- ❑ **Input:**
 - ❑ 1. SST-2:: Probe Training
 - ❑ 2. JinaAI Negation Dataset: Probe Training
 - ❑ 3. Neg-150-Simple: Intervention

Brain Scan (Probing)

- ❑ **Goal:** Map negation locations
- ❑ **Method:** Layer-wise linear probes
- ❑ **Measure:** Accuracy and AUCROC per layer/task

Diagnosis (Transfer Test)

- ❑ **Goal:** Validate generalization and trace knowledge
- ❑ **Method:** Transfer evaluation on JinaAI dataset and Representation Analysis
- ❑ **Measure:** Flip accuracy after patching

Surgery (Causal Interventions)

- ❑ **Goal:** Test bottleneck vs specialization
- ❑ **Method:** Cross-layer activation patching
- ❑ **Measure:** Flip accuracy after patching



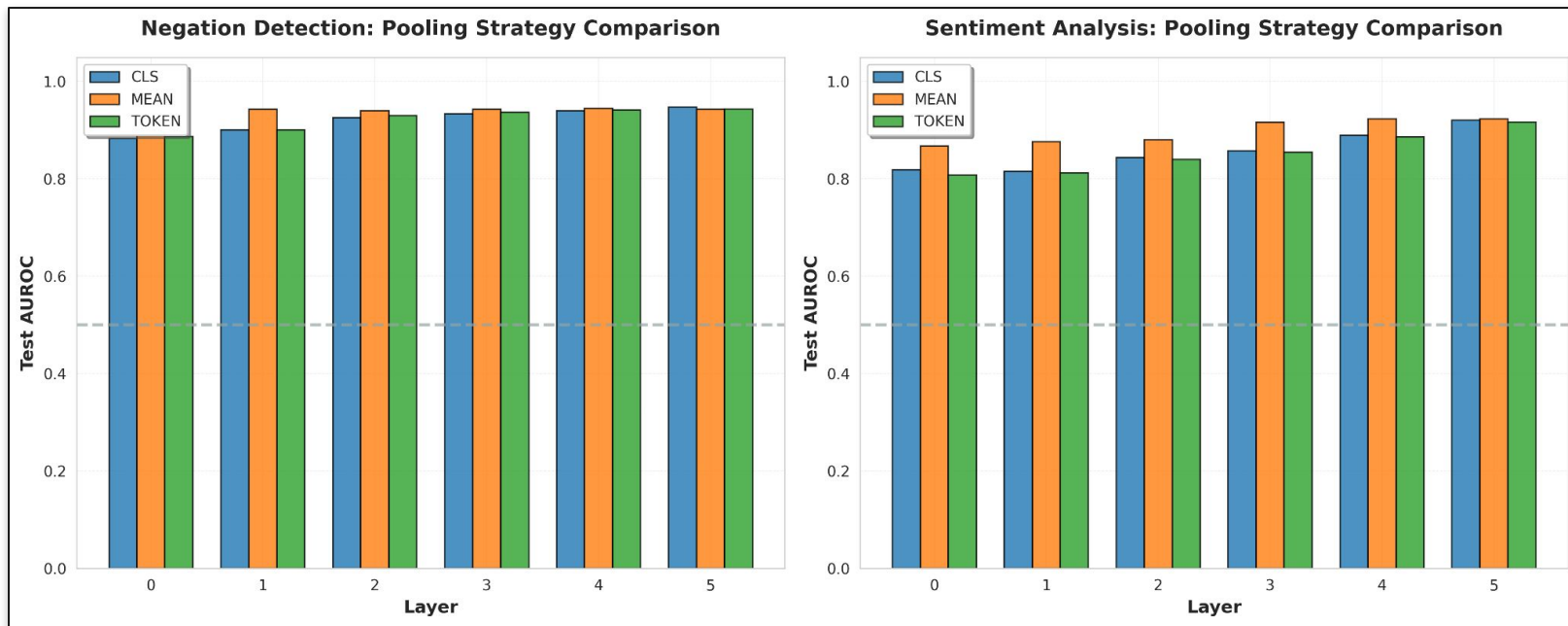
Datasets

Dataset	Sentence Count	Negated vs Non-negated	Purpose
SST-2 Base	10k train/2k dev/2k test	Mostly non-negated	Baseline sentiment capture
JinaAI	10k train/1k test	Mixed	Transfer Evaluation
NEG-150-SIMPLE	150 pairs	Mixed	Causal Testing

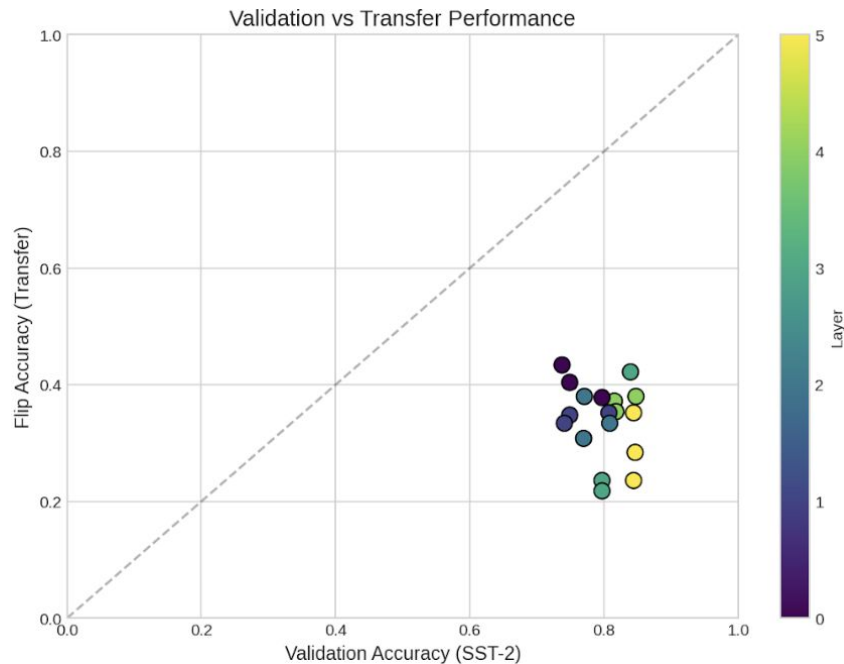
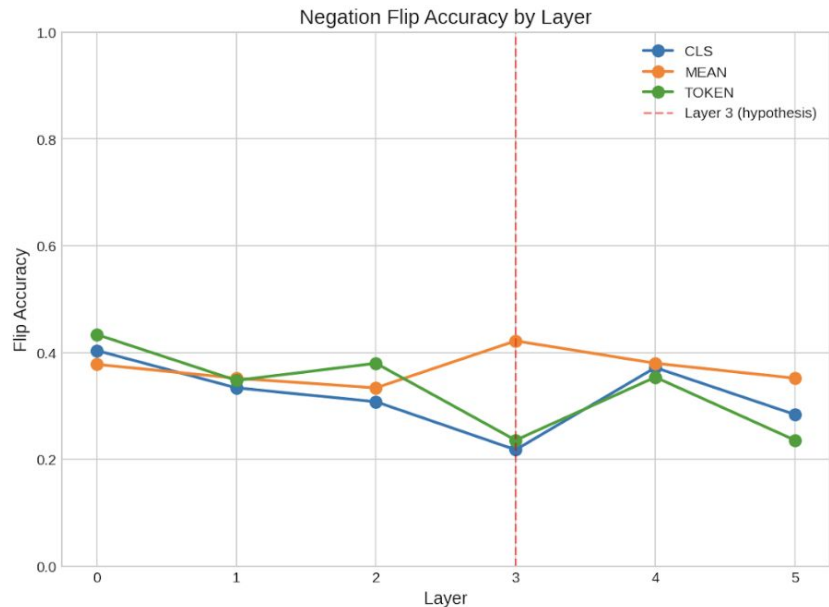


Results

Determine Whether Linear Probes Learned Well

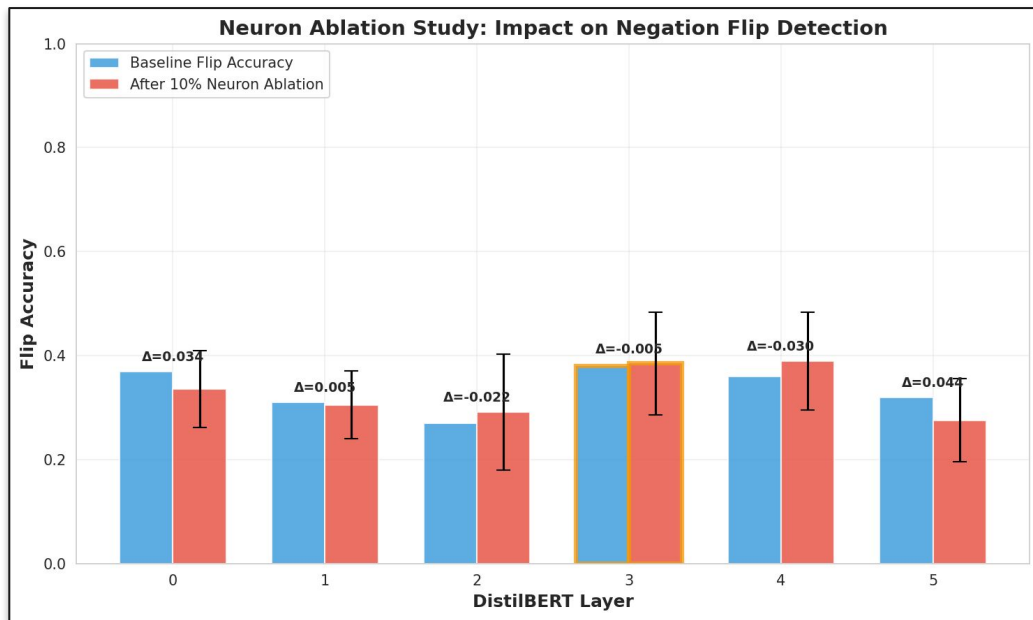


Seeing if this became generalized through a different task



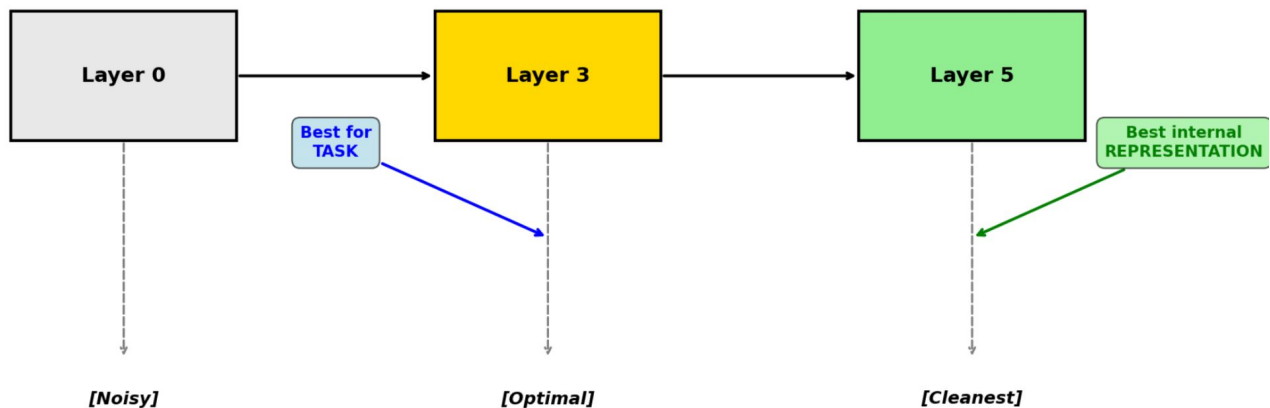
Why?

Layer 3 has the best results when it comes to Negation Decisions



Conclusion

KNOWLEDGE PROGRESSION THROUGH LAYERS:



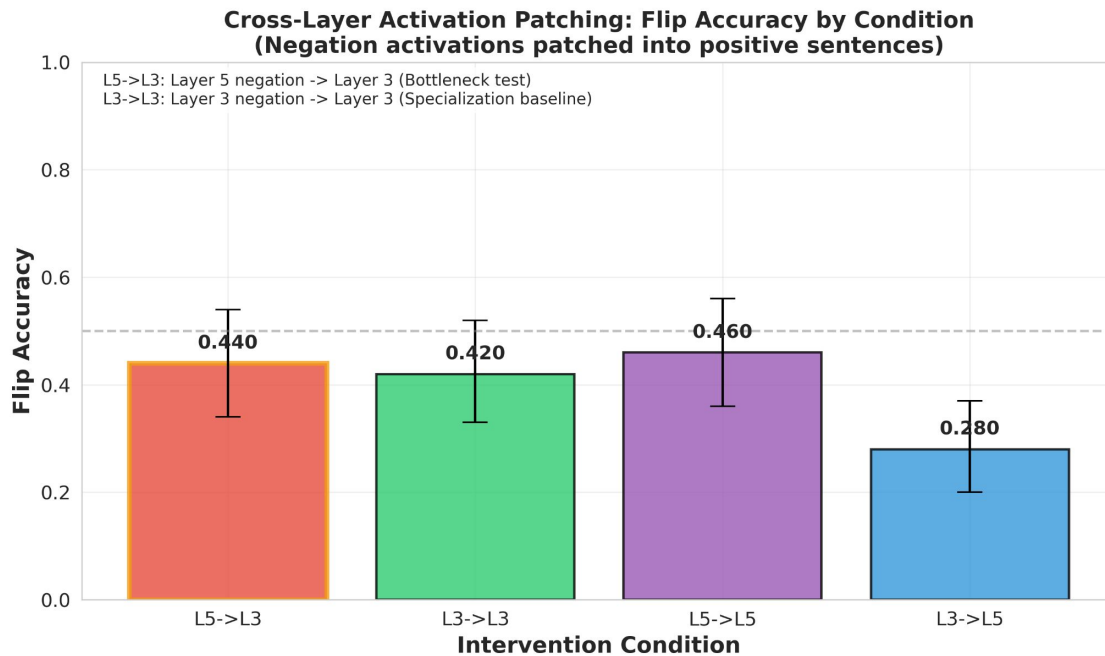
(but not matching!)



Critical Findings: Functional Split

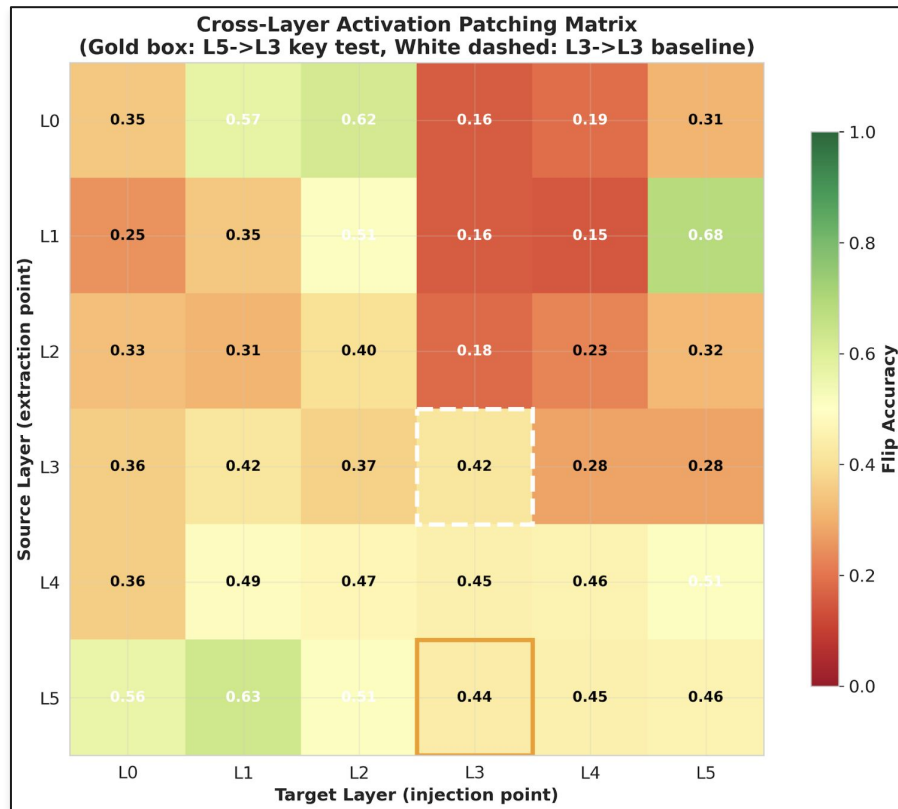
- Because of the AUROC/Accuracy at layer 5 and other cosine similarity metrics and layer 3 having the highest flip accuracy and confidence interval we can do patching

Activation Patching (“Swap” Test)



Targeted Ablations

- ❑ **Axis Logic:** Y-axis represents **Source Layer** ("clean" negation signal), and X-axis is the **Target Layer** ("injection point").
- ❑ **Hotspot (L5->L3):**
- ❑ **Interpretation:** Confirms casual hypothesis. Manually patching superior knowledge into decision making process.
 - ❑ Leads to a massive spike in label flipping





Results: “Decision vs. Representation” Gap

The Mechanism (Where is it?)

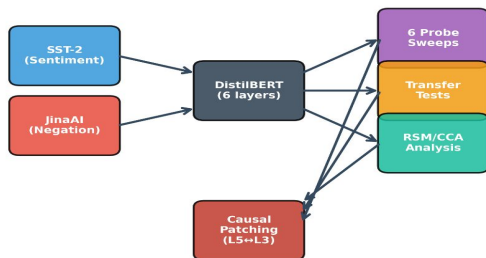
- ❑ Layer 3 is the “**decision maker**” for sentiment and negation.
 - ❑ Probing, transfer tests, and interventions confirm this
 - ❑ Influential layer as it directly impacts predictions (“positive” or “negative”)
- ❑ Layer 5 is the “**knowledge holder**” (The bottleneck)
 - ❑ Contains the most linearly separable negation features
 - ❑ **Insight:** Model possess refined negation logic deep in the network, but fails to leverage them earlier

The Process (How did it get there?)

- ❑ Evidence of Compression with BERT negation logic mapping to DistilBERT
 - ❑ The “student” model packs the “teacher’s” circuitry into denser earlier space
 - ❑ Upper negation → Middle band
- ❑ Causal Patching as a means to repair
 - ❑ Wiring the superior latent knowledge (L5) into its decision process (L3) to **fix negation errors**.
 - ❑ Effectively steer the model to fix negation errors and bypass bottleneck

Overview

A. Methodology Pipeline



Data → Model → Analysis → Causal Validation

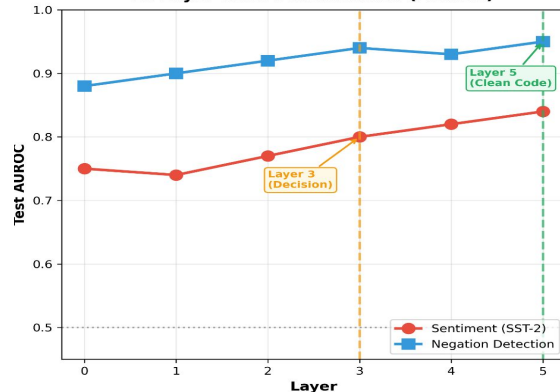
Approach

Layer
Comparison

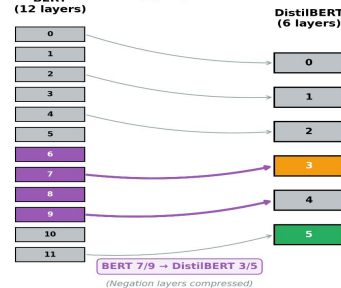
Feature
Compression

Proof

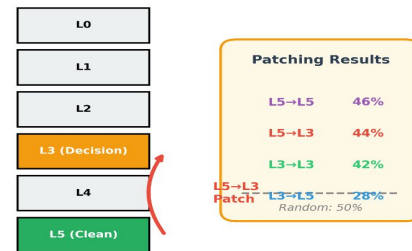
B. Layer-wise Performance (AUROC)



C. Distillation Mapping (BERT → DistilBERT)



D. Cross-Layer Causal Patching



Key: Knowledge works best where it lives (L5)



Conclusion

- ❑ **The Functional Split:** A critical gap exists between Decision (Layer 3) and Representation (Layer 5).
- ❑ **Compression Artifact:** Distillation forces complex logic into a dense "middle band," creating a decision bottleneck.
- ❑ **Steerability:** Cross-Layer Patching proves we can "rewire" the bottleneck to correct errors.

Future Work:

- ❑ **Verify generality:** Does this scale to BeRT model and does it affect other logical operators?
- ❑ **Explore Fixes:** Can we find other architectural fixes or new distillation techniques to fix this?
- ❑ **Practical Impact:** Creating a framework for detecting this gap in usage and existence in knowledge

Questions?
